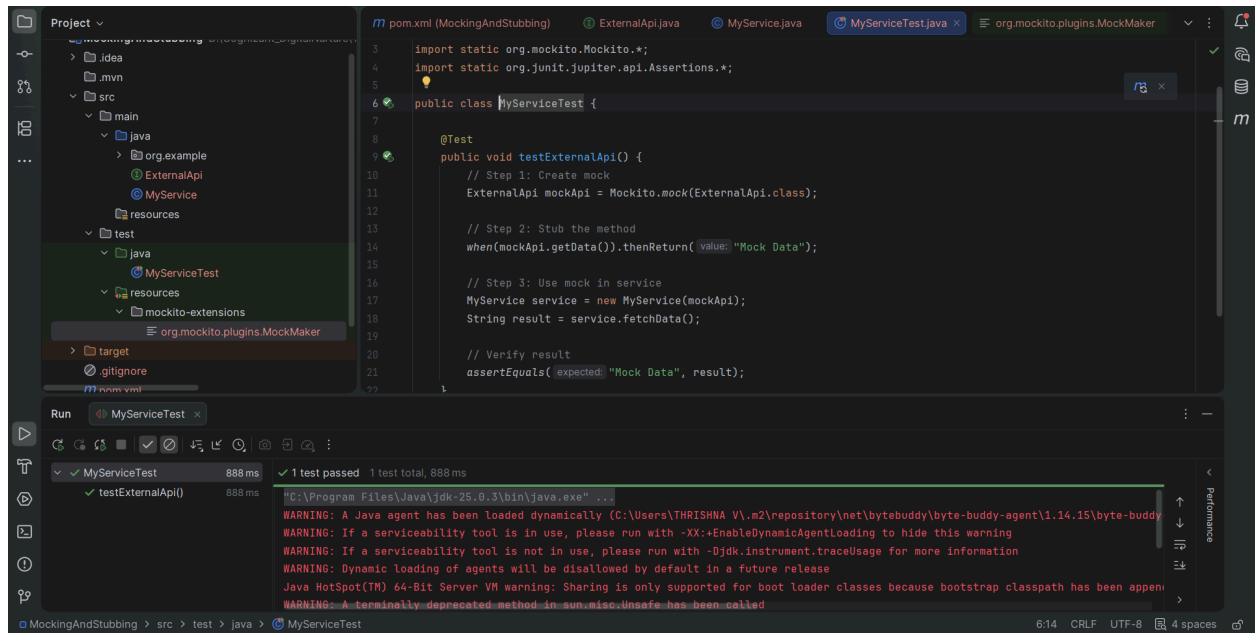


TDD using JUnit5 and Mockito

3. Mockito exercises

Exercise 1: Mocking and Stubbing



```
1 pom.xml (MockingAndStubbing)
2 ExternalApi.java
3 MyService.java
4 MyServiceTest.java
5 org.mockito.plugins.MockMaker

6 public class MyServiceTest {
7
8     @Test
9     public void testExternalApi() {
10         // Step 1: Create mock
11         ExternalApi mockApi = Mockito.mock(ExternalApi.class);
12
13         // Step 2: Stub the method
14         when(mockApi.getData()).thenReturn("Mock Data");
15
16         // Step 3: Use mock in service
17         MyService service = new MyService(mockApi);
18         String result = service.fetchData();
19
20         // Verify result
21         assertEquals("expected: 'Mock Data', result");
22     }
23 }
```

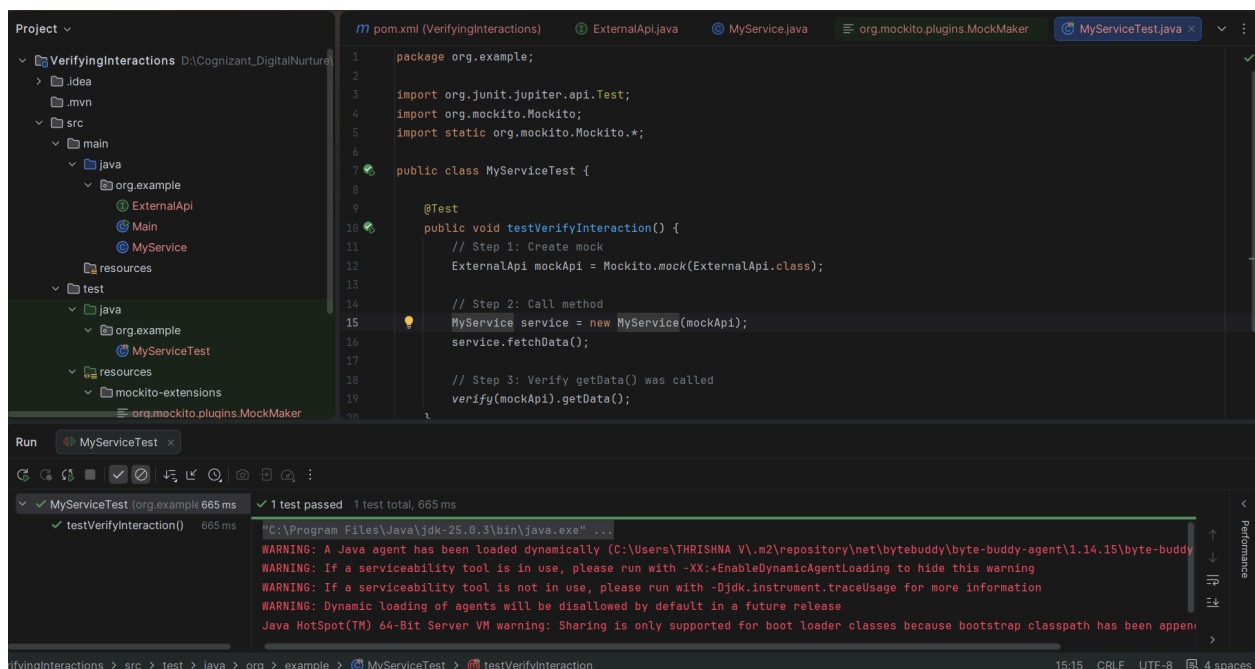
Run MyServiceTest

1 test passed 1 test total, 888 ms

testExternalApi() 888 ms

WARNING: A Java agent has been loaded dynamically (C:\Users\THRISHNA V\m2repository\net\bytebuddy\byte-buddy-agent\1.14.15\byte-buddy-agent-1.14.15.jar) but it is not in the bootstrap classpath. This warning can be ignored, but it is recommended to add the agent to the bootstrap classpath to avoid potential issues.

Exercise 2: Verifying Interactions



```
1 pom.xml (VerifyingInteractions)
2 ExternalApi.java
3 MyService.java
4 org.mockito.plugins.MockMaker
5 MyServiceTest.java

6 package org.example;
7
8 import org.junit.jupiter.api.Test;
9 import org.mockito.Mockito;
10 import static org.mockito.Mockito.*;
11
12 public class MyServiceTest {
13
14     @Test
15     public void testVerifyInteraction() {
16         // Step 1: Create mock
17         ExternalApi mockApi = Mockito.mock(ExternalApi.class);
18
19         // Step 2: Call method
20         MyService service = new MyService(mockApi);
21         service.fetchData();
22
23         // Step 3: Verify fetchData() was called
24         verify(mockApi).getData();
25     }
26 }
```

Run MyServiceTest

1 test passed 1 test total, 665 ms

testVerifyInteraction() 665 ms

WARNING: A Java agent has been loaded dynamically (C:\Users\THRISHNA V\m2repository\net\bytebuddy\byte-buddy-agent\1.14.15\byte-buddy-agent-1.14.15.jar) but it is not in the bootstrap classpath. This warning can be ignored, but it is recommended to add the agent to the bootstrap classpath to avoid potential issues.